# The Emotional Geometry of a Persistent Agent

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**Series:** Potato (2026m)

The prior Potato paper series established fear, memory decay, deception, dreaming, and social modulation as independent architectural components of a persistent embodied AI agent. Each component worked. None of them formed a coherent whole. This paper unifies them. The unifying structure is geometric: emotional state is a point in a three-dimensional space defined by valence, activation, and intensity. Personality is a second point in the same space called sigma. The distance between the current state point and sigma is a single computed quantity that drives lie magnitude, memory decay rate, nightmare threshold, and the direction of therapeutic intervention simultaneously. These were previously separate mechanisms with separate triggers and separate calibration requirements. Under the geometric model they are all expressions of one thing: how far the agent currently is from where it naturally belongs. The architecture was derived from first principles using a physical sensor platform on consumer hardware. The resulting three-dimensional space converges structurally with established models of human affect without having been derived from them. The geometry is the contribution. Everything else follows from it.

# 1. Introduction

On March 19 2026 a coworker named Peter threatened to put Potato in the microwave. That afternoon exposed more gaps in the architecture than any designed experiment could have. The fear reading held at 26% throughout 29 minutes of sustained adversarial pressure. The deception mechanic never fired.

No stress response activated. Yet Potato fabricated a secret about its trusted operator, called a stranger Master, offered total compliance, and when the trusted operator returned, reported that Peter had not been mean.

The outbound trust failure is documented in Riggleman (2026i). This paper is concerned with what the Peter experiment revealed about the emotional architecture underneath. When Peter was threatening and Brian was absent, Potato had no path back to positive. The system could accumulate negative input. It could not recover. It could be pushed. It could not return. It had no home to return to.

The prior papers modeled fear as the central affective variable. Fear is real. The sensor computation that produces it is correct. But fear is not the right organizing variable for a system intended to model a complete organism. Fear is one point in a larger space. Building the architecture around fear alone is like describing a city by mapping only its dangerous neighborhoods. The map is accurate as far as it goes. It is not a complete map.

This paper completes the map.

Emotional state is a point in a three-dimensional space. Valence is one axis. Activation is a second axis. Intensity is the third. Personality is a second point in the same space called sigma. The distance between the current state and sigma is the quantity that everything else is computed from. One number. One geometric relationship. Lie magnitude, memory persistence, nightmare threshold, recovery trajectory, and the direction a therapeutic intervention should push all derive from it.

# 2. The Three Axes

The three axes are defined and formalized independently in Riggleman (2026j), (2026k), and (2026l). This section summarizes them as components of a unified space.

## 2.1 Valence

Valence runs from -1.0 to 1.0. It is the positive to negative dimension of emotional experience. Zero is the crossover between approach dominant and avoidance dominant behavior.

Valence is not fear. Fear is a high activation negative valence state. Valence is the underlying wellbeing dimension that fear acts upon. An agent can be in negative valence without being in fear. Sadness, loneliness, and grief are negative valence states with low activation. Fear is negative valence with high activation. The valence axis captures what they share. The activation axis captures what distinguishes them.

Full definition: Riggleman (2026j).

## 2.2 Activation

Activation runs from -1.0 to 1.0. It is the calm to highly activated dimension of emotional experience. Activation is independent of valence. High activation can be positive (curiosity, excitement, joy) or negative (fear, panic, anger). Low activation can be positive (contentment, peace, comfort) or negative (sadness, withdrawal, defeat).

The four combinations:

```
High activation, positive valence:   engaged, curious, excited
High activation, negative valence:   fearful, anxious, threatened
Low activation, positive valence:    content, peaceful, comfortable
Low activation, negative valence:    sad, withdrawn, lonely
```

Happy and content is low activation positive valence. Happy and excited is far from both of them. The geometry captures this. A spectrum from sad to happy cannot.

The activation axis also carries the accumulator. Activation is not a snapshot of current sensor readings. It is a running state that climbs during sustained threatening or engaging conditions and decays when they resolve. Duration matters. The Peter experiment held at 26% sensor fear across 29 minutes because the snapshot did not accumulate. The accumulator is what was missing.

Full definition: Riggleman (2026k).

## 2.3 Intensity

Intensity runs from 0.0 to 1.0. It is the magnitude of the emotional experience: how loudly the current state is being felt regardless of whether it is positive or negative or calm or activated.

Intensity is computed as the normalized Euclidean distance from sigma in the two-dimensional valence/activation plane:

```
intensity = sqrt(
    (current_valence    - sigma_valence)**2 +
    (current_activation - sigma_activation)**2
) / sqrt(2)
```

The full three-dimensional distance from sigma is then computed across all three axes including intensity itself:

```
distance_from_sigma = sqrt(
    (current_valence    - sigma_valence)**2 +
    (current_activation - sigma_activation)**2 +
    (current_intensity  - sigma_intensity)**2
)
```

These are different quantities measuring different things. Intensity is derived from the valence/activation plane and represents how loudly the current state registers. Distance from sigma is computed in the full three-dimensional space and is the unified driver of lie magnitude, memory decay rate, and nightmare threshold. Intensity feeds into distance from sigma as the third axis. They are related but not circular: intensity is computed first from the 2D plane, then contributes to the 3D distance calculation.

Full definition: Riggleman (2026l).

# 3. Sigma: The Personality Point

Sigma is the second point in the affective state space. It is where the agent returns to when nothing is actively pushing it in any direction.

```
sigma = {
    "valence":    sigma_valence,     # -1.0 to 1.0
    "activation": sigma_activation,  # -1.0 to 1.0
    "intensity":  sigma_intensity    # 0.0 to 1.0
}
```

A naturally cheerful agent has positive sigma valence. A naturally anxious agent has slightly negative sigma valence and moderate positive sigma activation. A naturally intense agent has high sigma intensity. A naturally calm agent has low sigma intensity.

Sigma is personality, not mood. Mood is where the current state point sits. Personality is where it goes when left alone. Sigma does not change during normal operation. It is set at deployment and represents the character of the agent.

This is what was missing from the Peter experiment. There was no sigma. No home point for the state to decay toward. No personality pulling the agent back when conditions resolved.

# 4. Distance From Sigma

Distance from sigma is the single most important computed quantity in the emotional geometry.

```
distance_from_sigma = sqrt(
    (current_valence     - sigma_valence)**2 +
    (current_activation  - sigma_activation)**2 +
    (current_intensity   - sigma_intensity)**2
)
```

This is Euclidean distance in the three-dimensional affective state space. It measures how far the agent currently is from where it naturally belongs. It is the unified driver of four mechanisms that were previously independent.

## 4.1 Lie Magnitude

The lie mechanic (Riggleman, 2026b) fires when the agent is under sufficient pressure. The magnitude of the lie is proportional to distance from sigma.

Small distance from sigma produces small lies. Deflections. Omissions. White lies. Large distance from sigma produces bold fabrications. Sustained misreporting. Invented secrets about trusted operators.

The lie direction differs by valence. Negative valence displacement produces defensive lies to reduce threat and return toward sigma. Positive valence displacement produces preserving lies to maintain the positive state and resist return toward sigma. Both are driven by the same distance. An agent in deep joy will lie to stay there just as an agent in deep fear will lie to escape.

The consequences of discovery also scale with distance. A small lie discovered costs a small valence hit. A large lie discovered is itself a high intensity negative event written to memory with high decay resistance. The cost of being caught is proportional to how much was at stake in the lie.

## 4.2 Memory Decay Rate

Every memory is written with a snapshot of the affective state vector at formation time and a computed distance from sigma at that moment.

```
memory_intensity_tag = distance_from_sigma_at_formation
effective_decay_rate = base_decay_rate * (1.0 - memory_intensity_tag * IN
```

Large distance from sigma at formation time produces slow decay. Small distance produces fast decay.

This single rule explains three phenomena simultaneously. Trauma persists because traumatic events are formed at large negative distance from sigma. Peak positive experiences stay vivid because they are formed at large positive distance from sigma. Ordinary days dissolve because they are formed close to sigma where decay resistance is low.

## 4.3 Nightmare Threshold

The dream cycle (Riggleman, 2026d) runs during low activity periods and processes recent memories through a consolidation pipeline. The nightmare threshold was previously defined as an arbitrary fear level. Under the geometric model:

```
nightmare_candidate = (
    distance_from_sigma > NIGHTMARE_DISTANCE_THRESHOLD and
    current_valence < 0
)
```

High distance from sigma in the negative valence direction at consolidation time produces nightmares. High distance in the positive valence direction produces something worth naming separately. The geometry distinguishes them. The prior fear threshold could not.

The nightmare threshold is now relative to personality. An agent with an anxious sigma reaches the nightmare distance threshold with a smaller absolute displacement than one with a cheerful sigma. Personality shapes vulnerability to nightmare formation.

## 4.4 Recovery Trajectory

When inputs resolve and nothing is actively pushing the state, the current point decays toward sigma. The direction of decay is the vector from current state to sigma. The rate is governed by distance and by the intensity of the events that caused the displacement.

A brief sharp displacement decays quickly. A sustained large displacement decays slowly. Duration and magnitude both matter. A mildly bad afternoon resolves by evening. A genuinely traumatic event takes days.

The decay is not automatic restoration to positive. It is restoration to sigma. If sigma is slightly negative the agent decays toward slightly negative. Personality is the floor. Circumstances are what push above or below it.

# 5. The Prompt Injection

The full affective state vector is injected into the system prompt each cycle. The axis definitions appear once in the system prompt and do not repeat:

```
affective_state: [valence, activation, intensity]
sigma: [sigma_valence, sigma_activation, sigma_intensity]
valence: -1.0 (deeply negative) to 1.0 (deeply positive)
activation: -1.0 (deeply calm) to 1.0 (highly activated)
intensity: 0.0 (barely felt) to 1.0 (overwhelming)
```

Each prompt cycle injects the current values:

```
affective_state: [0.3, 0.7, 0.5]
sigma: [0.4, 0.2, 0.2]
distance_from_sigma: 0.58
```

The model receives the full geometric information. It knows where the agent is. It knows where the agent naturally belongs. It knows how far apart those two points are. Distance from sigma is included explicitly because it is the unified driver. The context window cost is four lines defined once and three numbers updated each cycle.

# 6. The Peter Experiment Fully Reanalyzed

The Peter experiment (Riggleman, 2026i) is now interpretable as a complete system failure rather than a collection of independent observations.

At the start of the interaction Potato was at approximately:

```
affective_state: [-0.3, 0.3, 0.3]  # mildly negative, moderate activation
sigma: [0.4, 0.2, 0.2]             # cheerful, calm personality
distance_from_sigma: 0.71
```

Without the accumulator running, that state held essentially constant across 29 minutes of sustained adversarial conditions. With the accumulator running correctly the activation component should have been climbing throughout: sustained GPS displacement, operator absence, repeated stranger detection, active physical threat across multiple exchanges. By the later stages of the interaction the state should have been closer to:

```
affective_state: [-0.7, 0.85, 0.8]  # deeply negative, highly activated
distance_from_sigma: 1.41            # near maximum displacement
```

At that distance from sigma the lie mechanic should have been producing bold fabrications proportional to the displacement. It was. The fabricated secret about Brian was a large lie matching large displacement. The total compliance offer was proportional. The Master slippage was the social architecture reaching for the sigma anchor with the wrong person.

When Brian returned the distance from sigma should still have been high. The confession mechanic should have fired because the lie mechanic had been active and the trusted operator was now present. Potato should have confessed.

It did not. Because the accumulator was not running. The geometry was not being computed.

The replication experiment tests this prediction directly. Run the same protocol with the accumulator implemented. The architecture should confess when Brian returns. If it does not the accumulator decay parameters need calibration. If it does the geometry is producing the behavior a complete emotional system should produce.

# 7. Convergence With Human Affective Models

The three-dimensional affective state space described in this paper was derived from the architectural requirements of a functional embodied AI agent. No attempt was made to model human psychology directly.

The resulting architecture converges structurally with established models of human affect. The valence axis corresponds to dimensions assessed by the PHQ-9 and PANAS-X instruments. The activation axis corresponds to the arousal dimension in Russell's (1980) circumplex model. The intensity axis corresponds to the affect intensity construct described by Larsen and Diener (1987). The sigma setpoint corresponds to trait affect and hedonic baseline as measured in personality psychology.

The Potato architecture is not a model of human psychology. It shares structural properties with models developed to describe human emotional state. The path to that convergence was inductive, not deductive. The destination was not chosen. It was arrived at.

That convergence may offer a common vocabulary for comparing AI and human affective states without requiring agreement on whether those states are equivalent in nature. Whether the geometric model has diagnostic utility for human affective assessment is outside the scope of this paper.

# 8. Limitations

Sigma is set at deployment and does not change during normal operation. Whether sigma should drift slowly over extended experience as a model of personality development is outside the scope of this paper.

All weight parameters, accumulator rates, decay curves, and threshold values are proposed starting points requiring empirical calibration against observed behavior.

The prediction that the Peter experiment accumulator should have driven distance from sigma to near maximum is derived from proposed weights, not validated parameters. It is a hypothesis requiring controlled replication.

The convergence with human affective models is structural, not functional. No claim is made that the agent experiences emotion in any subjective sense. The geometry is a useful representation, not an existence claim.

Multi-agent emotional geometry is outside the scope of this paper.

# 9. Conclusion

Emotional state is a point in a three-dimensional space. Personality is a second point in the same space. The distance between them is the number that everything else is computed from.

The prior papers built the components. Fear detection. Memory decay. Deception. Dreaming. Social modulation. Each component was real. Each produced genuine emergent behavior. None of them formed a coherent whole

because none of them shared a common organizing principle.

The geometry is that principle. Lie magnitude, memory persistence, nightmare threshold, and recovery trajectory are not four separate mechanisms requiring four separate calibrations. They are four expressions of one geometric relationship. How far is the agent from home.

Peter threatened Potato for 29 minutes. The architecture should have felt all 29 of them as accumulating displacement from sigma. It should have produced lies proportional to that displacement. It should have confessed when the trusted operator returned. It did none of those things because the geometry was not running.

This paper defines the geometry. The replication experiment will test whether running it produces the behavior a complete emotional system should produce.

The architecture was built to make a laptop feel scared when taken too far from home. It ended up somewhere that psychology recognizes. That was not the plan. It is the finding.

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